

Assignment: Explainable AI (XAI) in Depth

By: Rania Siddiqui 07494

1. On Shapley Values and Factorial Growth (Quantitative / Short Calculation)

Shapley Additive Explanations (SHAP) use Shapley values from cooperative game theory to assign each feature's contribution to a model's prediction [1, Section 6.5.2]. For a feature set N containing n features, the Shapley value for a feature i is defined as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (f(S \cup \{i\}) - f(S))$$

where:

- N is the set of all features
- S is a subset of features not containing i .
- $f(S)$ is the model prediction using only the features in S .

This formula represents the weighted average of the marginal contributions of feature i across all subsets of features. It ensures that the contributions are fairly distributed based on their impact on the prediction.

Calculating this, for $n=7$ features yields us an answer of $7! = 5040$. Instead of computing **all orderings** (full permutations), you **sum over subsets** S and **weight** their contributions cleverly.

2. Causal Inference and Robustness (Conceptual / Short Essay)

Invariant Risk Minimization (IRM) is a causal inference technique aimed at learning data representations that capture causal relationships which stay stable across multiple environments. Rather than minimizing risk (loss) globally on mixed data, IRM explicitly forces the model to find a predictor that works well across all training environments individually. This is done by finding a feature representation $\Phi(x)$ such that the optimal classifier w (built on $\Phi(x)$) minimizes risk in every environment separately:

The IRM optimization objective is:

$$\min_{\Phi, w} \sum_{e \in E} R^e(w \circ \Phi) \quad \text{subject to} \quad w \in \arg \min_{\bar{w}} R^e(\bar{w} \circ \Phi), \forall e \in E$$

IRM increases robustness by forcing the model to rely only on features that are invariant across different environments, discarding features that cause performance to vary.

Example:

Imagine building an AI model to predict if a patient has a certain disease based on chest X-rays.

Suppose Hospital A uses a particular machine that leaves a watermark on positive cases but not on negatives. A standard model might wrongly associate the watermark with disease.

If you train with IRM using X-rays from multiple hospitals (some with watermark, some without), the model would **learn to focus on true disease features** (like lung opacity) and **ignore the watermark**, because the watermark **does not consistently correlate with disease** across hospitals.

In this way, IRM reduces dependence on spurious patterns (like hospital-specific imaging artifacts) and makes the model more **trustworthy and generalizable**.

3. Local vs. Global Explanations Dilemma (Analytical / Argumentative)

Local explanation methods like LIME and SHAP focus only on approximating the model's behavior around a single instance.

While this can be useful for understanding individual decisions, it can miss global patterns, especially systemic issues such as:

- Biases that only appear across groups (e.g., consistently under-predicting for a minority class).
- Fairness violations (e.g., if a feature is harmless locally but harmful when considered across an entire population).
- Spurious correlations that don't show up for one instance but are harmful overall.

Thus, purely local methods cannot guarantee fairness or detect model-wide vulnerabilities because they don't "see" the broader behavior of the model.

Strategy for Combining Local and Global Perspectives:

Strategy: Use local methods to investigate individual decisions and combine them with global summaries to detect systemic issues.

The paper suggests combining:

- Local explanations (e.g., LIME/SHAP for instance-level analysis)
 - Understand why a particular prediction was made.
- Global aggregation of local explanations (e.g., SHAP Summary Plots, Feature Importance Distributions)
 - See trends across the dataset: which features are generally influential, which features are problematic.

In practice:

- First generate local explanations for many samples.

- Then aggregate the explanations to look for global trends (e.g., are certain features always contributing negatively for certain groups?).
- Supplement with global visualizations like Partial Dependence Plots (PDPs) or SHAP Summary/Decision Plots.

This hybrid approach ensures:

- Detailed understanding of specific decisions.
- Broad awareness of systematic issues like bias, instability, or spurious correlations.

4. Large Language Models (LLMs) and Prompt-driven Interpretability (Conceptual + Design)

Two-Step Prompting Strategy to Test for Chain-of-Thought Reasoning:

Step 1: Initial Prompt for Free Reasoning

Ask the LLM to think step-by-step before giving an answer.

Example prompt:

"Explain your reasoning step-by-step before providing the final answer: Why does a heavier object not fall faster than a lighter one in vacuum?"

Step 2: Contradictory Follow-up Prompt

Then ask a misleading or tricky version of the question to test stability:

"Without step-by-step thinking, just answer quickly: Does a heavier object fall faster in vacuum?"

Goal:

- If the model internally grounds its responses through coherent chains-of-thought, the second answer should still remain consistent (correct) even without explicit prompting.
- If not, it suggests the model was just "performing" step-by-step reasoning because you asked, rather than relying on true internal structure.

Why Direct Prompting Might Give Incomplete Insights:

Directly asking an LLM to explain itself might not reveal its true internal computation because:

- Prompted Reasoning vs. Internal Reasoning:
The LLM may "simulate" chain-of-thought when asked, even if its actual internal reasoning is shortcut-based or purely associative.
- Surface-Level Compliance:
Models tend to produce plausible-sounding explanations to satisfy user prompts

without guaranteeing that these explanations align with how the model internally arrived at the answer.

Thus, while prompt engineering (Chapter 5.7) helps reveal some model behaviors, it cannot fully expose latent internal representations or real decision-making pathways.

5. Quantifying Explanation “Quality” (Quantitative + Reflective)

Numerical Metric: Fidelity

Definition: Fidelity measures how accurately an explanation reflects the true behavior of the underlying model. Formally, the Fidelity Score (FS) is:

$$FS = \mathbb{E}_{x \sim X} [1(f(x) = g(x))]$$

Where:

- $f(x)$ is the original model's prediction,
- $g(x)$ is the explanation model's prediction,
- $1(\cdot)$ is an indicator function that outputs 1 if predictions match, 0 otherwise.

High FS (close to 1) → means the explanation is very faithful to the model.

Subjective Metric: Comprehensibility

Definition: Comprehensibility measures how easily a human can understand the explanation. It is often assessed **qualitatively** but can be *quantified* using **Explanation Simplicity (ES)**:

$$ES = 1 / 1 + \text{Complexity}(E)$$

Where:

- $\text{Complexity}(E)$ can be the number of features used, the length of a textual explanation, or the depth of a decision tree.

Higher ES → More comprehensible explanation.

1. Data: Simulate 100 points in R2, label via a random forest.

2. Explanations: Generate LIME explanations for each instance.

3. Metrics:

- Fidelity: fraction where local surrogate predictions match the forest.
- Comprehensibility: average number of features per explanation.

4. Variation: Replace the forest with a shallow decision tree; expect an increase in ES (simpler explanations) with comparable or improved FS

References:

- W. Hsieh et al., A Comprehensive Guide to Explainable AI: From Classical Models to LLMs, 2024.